LAB 05

Indicator Stage: Pine Script v6 Indicators and their Python Twins

Item	Detail
Topic	Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca
Pipeline stage	Indicators
TSC reference	A3, K3, K7
Duration	25 min
Folder	labs/Lab05-pine-indicators/
Files	Lab05-pine-indicators/indicators_stage.py, Lab05-pine-indicators/lcc_trend_bands.pine, Lab05-pine-indicators/lcc_rsi.pine, Lab05-pine-indicators/tradingview_indicator_values.csv, pipeline/indicators.py, Lab05-pine-indicators/solution/indicators.py
Simulator page	http://localhost:8000/#/data?tab=chart&symbol=SPY&overlays=sma:50,sma:200,bollinger
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital's traders read trend (EMA 20/50), volatility (Bollinger Bands) and momentum (RSI) on TradingView, while the pipeline's signals are computed in Python. If the two disagree, a trader and the automated strategy will act on different numbers, so the desk builds Pine Script indicators and exact Python twins and cross-checks them value by value.

Objective

Add the desk's Pine Script v6 indicators to a TradingView chart, build pipeline/indicators.py with twins of ta.ema, ta.sma, ta.rsi, ta.bb, ta.atr, ta.macd, ta.crossover and ta.crossunder, and cross-check the values against the TradingView Data Window.

Before You Start

- Part A: start the Trading Simulator (uv run trader serve --with-worker, or docker compose up -d) and open http://localhost:8000 > Course Labs > Lab 05. Install steps: Learner Guide, 'Download, Install and Start the Trading Simulator Software'.
- Part B: open a terminal in the labs folder, activate your virtual environment and open your AI coding assistant there; the finished reference code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

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Part A - In the Trading Simulator

Start the simulator (uv run trader serve --with-worker, or docker compose up -d) and open http://localhost:8000. In the sidebar open Course Labs > Lab 05: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

A1. Overlay trend indicators

Data > Chart for SPY: toggle sma:50, sma:200, ema:20 and bollinger.

Open: http://localhost:8000/#/data?tab=chart&symbol=SPY&overlays=sma:50,sma:200,bollinger (Data > Chart SPY with SMA 50/200 and Bollinger)

Check: Lines draw over the candles (TradingView Lightweight Charts).

A2. Oscillators

Switch the lower pane between RSI(14) and MACD.

Open: http://localhost:8000/#/data?tab=chart&symbol=SPY&osc=macd (Data > Chart SPY MACD)

Check: RSI has 30/70 guide lines; the MACD histogram changes colour above and below 0.

A3. Export Pine Script v6

Strategy Library > EMA crossover > Pine Script v6 > Copy. In TradingView open the Pine Editor, paste, 'Add to chart'.

Open: http://localhost:8000/#/strategies?key=ema_cross (Strategies > EMA crossover)

Check: TradingView shows the two EMAs; their last values match the app's chart.

Expected in the app: A chart with overlays and an oscillator pane, and a Pine Script pasted into TradingView whose values match.

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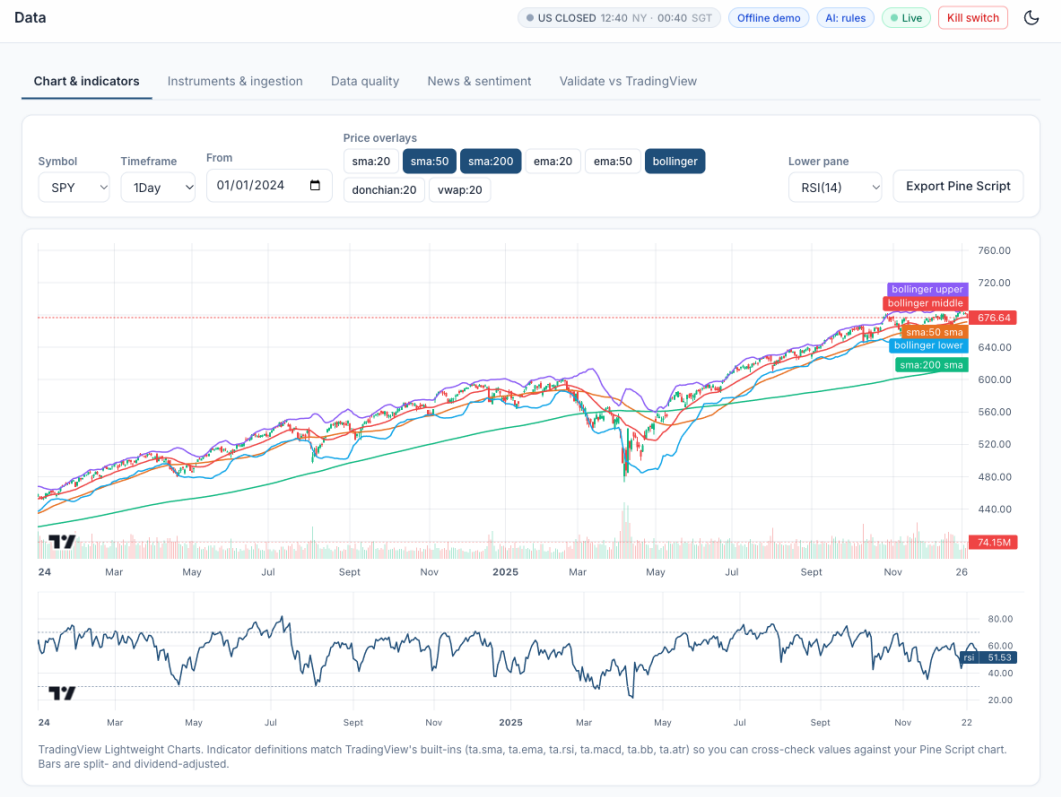


Figure: Lab 05 in the Trading Simulator (Lab05_indicators_pine.png).

Part B - In Code with Your AI Assistant

The same pipeline stage in code. AI writes the code; you own the algorithm: paste each prompt into your AI coding assistant in the labs folder (the finished reference code is already in labs/Lab05-pine-indicators/), run it, and check the result against Part A. The Practical Performance assessment uses this code path.

Step 1: Add the trend and bands indicator (A3, K3)

On TradingView open NASDAQ:AAPL, 1D. Open the Pine Editor, create a new indicator, replace the template with Lab05-pine-indicators/lcc_trend_bands.pine (it starts with //@version=6), Save as 'LCC Trend' and Add to chart.

AI prompt - paste into your AI coding assistant
Explain labs/Lab05-pine-indicators/lcc_trend_bands.pine line by line for a finance professional: the inputs, ta.ema, ta.bb, ta.crossover / ta.crossunder, plotshape, barcolor and alertcondition. Which values will the Data Window show for each bar?

Check: EMA 20 (blue), EMA 50 (orange), shaded Bollinger Bands and green/red crossover triangles on the price pane.

Step 2: Add the RSI pane (A3, K3)

Create another indicator from Lab05-pine-indicators/lcc_rsi.pine, save it as 'LCC RSI' and add it.

Check: RSI(14) in its own pane with 70/30 lines and shaded overbought/oversold zones.

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Step 3: Record the TradingView values (A3)

Move the cursor to 31 December 2025, open the Data Window and copy close, EMA fast, EMA slow, BB upper, middle, lower and RSI into Lab05-pine-indicators/tradingview_indicator_values.csv (overwrite the example row).

Check: Default (unadjusted) chart: close 271.86, EMA 20 about 273.98, EMA 50 about 269.60, BB 282.35 / 275.47 / 268.60, RSI about 45.64.

Step 4: Build the Pine twins (K3)

Ask your assistant for the first half of pipeline/indicators.py.

AI prompt - paste into your AI coding assistant
Create labs/pipeline/indicators.py with Python twins of the Pine Script v6 built-ins, implemented by calling labs/common/indicators.py: ta_sma(src, length), ta_ema(src, length) (alpha 2/(n+1), adjust=False), ta_rsi(src, length=14) (Wilder smoothing), ta_bb(src, length=20, mult=2.0) returning (middle, upper, lower) with the population standard deviation, ta_atr(bars, length=14), ta_macd(src, 12, 26, 9) returning (macd, signal, hist), ta_crossover(a, b) = a > b and a[1] <= b[1], ta_crossunder(a, b). Then lcc_trend_bands(bars, fast=20, slow=50, bb_len=20, bb_mult=2.0) returning close, ema_fast, ema_slow, bb_middle, bb_upper, bb_lower, bull_cross, bear_cross exactly like lcc_trend_bands.pine, lcc_rsi(bars, 14, 70, 30) returning rsi, alert_oversold and alert_overbought like lcc_rsi.pine, and indicator_table(bars) joining both. Every value at bar t may use data up to bar t only. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: indicator_table(load_bars('AAPL','2018-01-01','2025-12-31')).iloc[-1] shows ema_fast 273.23 and rsi 45.69.

Step 5: Build the cross-check run (A3, K3)

Ask your assistant for run() and main().

AI prompt - paste into your AI coding assistant
In labs/pipeline/indicators.py add run(symbol='AAPL', date='2025-12-31', tv_file=None) for `python -m pipeline indicators --symbol --date --tv`: compute indicator_table on the ADJUSTED research bars (from config data.start) and on the RAW reference bars from pipeline.validate.reference_bars (TradingView's default view); print a snapshot table (close, ema_fast, ema_slow, bb_upper, bb_middle, bb_lower, rsi) with columns Python ADJUSTED, Python RAW (TV default), TradingView (you) from Lab05-pine-indicators/tradingview_indicator_values.csv and 'diff vs RAW' in %; the trend, band and RSI reading; the number of bullish/bearish crosses and RSI alert events since 2024-06-01; PASS if the largest gap is <= 0.5%. Save _outputs/Lab05_<SYMBOL>_indicators.csv and chart Lab05_pine_preview.png (price, EMAs, bands, cross markers, RSI pane). Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline indicators --help shows --symbol, --date and --tv.

Step 6: Run the cross-check (A3, K3)

Run: python -m pipeline indicators --symbol AAPL --date 2025-12-31 (same as: python3 Lab05-pine-indicators/indicators_stage.py --symbol AAPL).

Check: ADJUSTED close 271.12, EMA 20/50 273.23 / 268.80, BB 281.58 / 274.72 / 267.87, RSI 45.69; RAW 271.86, 273.98 / 269.60, 282.35 / 275.47 / 268.60, RSI 45.64; largest gap to TradingView 0.006% -> PASS; trend UP; since 2024-06-01: 2 bullish and 2 bearish EMA crosses, 3 RSI-oversold and 15 RSI-overbought alert events.

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Step 7: Change the inputs and count the extra trading (K3, K7)

In TradingView set LCC Trend to EMA 10/30 and watch the triangles multiply. Then measure it in Python.

AI prompt - paste into your AI coding assistant
In a Python shell started in labs, use pipeline.indicators.lcc_trend_bands on pipeline.ingest.load_bars('AAPL', '2018-01-01', '2025-12-31') with fast/slow 20/50 and then 10/30, count bull_cross and bear_cross since 2024-06-01 for each, and estimate the extra trading cost of the faster setting at 3 bps per round trip on USD 100,000. Print a small table.

Check: Since 2024-06-01: EMA 20/50 = 2 bullish / 2 bearish crosses; EMA 10/30 = 7 / 7 - about 3.5 times as many trades, so cost rises with signal frequency.

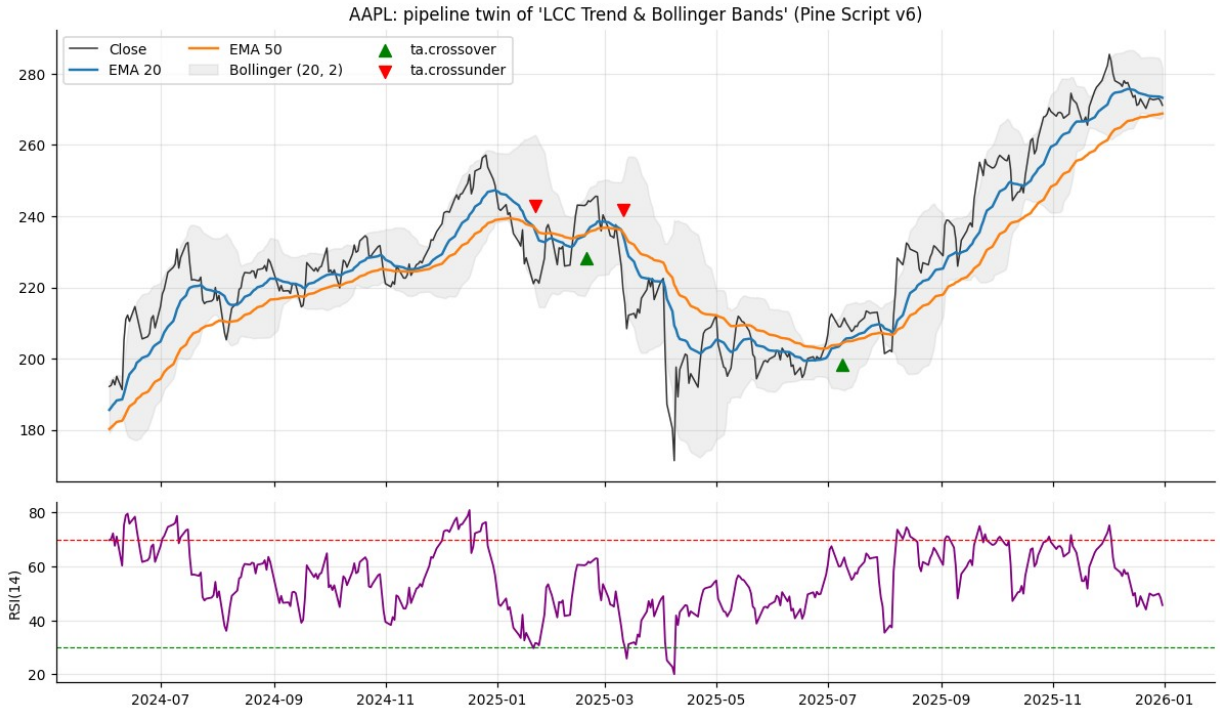
Step 8: Create the alert the automation stage will use (A3, K7)

Restore EMA 20/50. Click Alert, choose Condition: LCC Trend > 'EMA bullish cross', trigger 'Once per bar close', and save. This alert type is what Lab 17 connects to the pipeline webhook.

Check: The alert appears in the Alerts panel, active, on NASDAQ:AAPL.

Expected Output

python -m pipeline indicators --symbol AAPL --date 2025-12-31: adjusted close 271.12, EMA 20/50 273.23 / 268.80, Bollinger 281.58 / 274.72 / 267.87, RSI(14) 45.69; raw (TradingView default) 271.86, 273.98 / 269.60, 282.35 / 275.47 / 268.60, RSI 45.64; largest gap to TradingView 0.006% -> PASS; 2 bullish and 2 bearish EMA crosses since June 2024; files _outputs/Lab05_AAPL_indicators.csv and chart _outputs/Lab05_pine_preview.png.



Lab 5 output (Lab05_pine_preview.png), offline mode on the bundled data.

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Debrief

Technical indicators (K3) turn prices into trend, volatility and momentum readings; defining them once in Pine Script and once in Python, and proving they match (A3), keeps traders and the automated pipeline on the same numbers. Faster settings fire more signals, which means more trades and more cost (K7).

Troubleshooting

Symptom	What to do
Compilation error in the Pine Editor	The first line must be //@version=6 and the whole file must be pasted.
Indicator draws in a separate pane	lcc_trend_bands.pine must keep overlay = true; only lcc_rsi.pine uses overlay = false.
Gap to TradingView above 0.5%	Check the chart is 1D, 'Adjust data for dividends' is unticked, and you read the 2025-12-31 bar; compare with the RAW column.
FileNotFoundError for the RAW column	Raw reference files exist for AAPL and D05.SI only; other symbols show the adjusted column.

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